# Detecting the time-reversal-odd $A_2$ bilinear of a chiral superconductor with spin-polarized quasiparticle interference

## SUMMARY

1. A two-component order parameter in an irrep $E$ has bilinears $E \otimes E = (A_1 \oplus E)_s \oplus (A_2)_a$. The antisymmetric part $A_2$ is the chirality. It is odd under time reversal $\mathcal{T}$.

2. *Obstruction.* Spin-integrated STM around a non-magnetic impurity is $\mathcal{T}$-even. The impurity-dressed LDOS is therefore identical in the two chiral domains, to all orders in the impurity potential and for any impurity shape. The $A_2$ bilinear never reaches the signal.

3. *Resolution.* A spin-polarized tip measures the spin-resolved LDOS, which is $\mathcal{T}$-odd. With spin–orbit coupling (SOC), this signal vanishes for every $\mathcal{T}$-invariant superconductor and is strictly odd in the chirality. It is the impurity-resolved image of the $A_2$ bilinear.

4. *Classification.* The mirror parity of the spin-resolved map separates chiral states (antiunitary mirrors) from non-chiral $\mathcal{T}$-breaking states (unitary mirrors). The symmetry decomposition of the clean LDOS detects nematicity. Together they identify the pairing state and the symmetry of each defect.

5. *Comparison with Ref.*[1]. The node–antinode fingerprint detects that the gap belongs to a non-trivial irrep ($E$), but not how its two components are combined. Our protocol detects the $A_2$ bilinear itself, and hence chirality.

## I. SETUP

Write the order parameter as a doublet $(\Delta_x, \Delta_y)$ in an irrep $E$ (for Sn/Si(111), $E_2$ with $d$-wave form factors) and fix the global phase so that

$$\Delta(\mathbf{k}) = \Delta_x(\mathbf{k}) + i\chi\,\Delta_y(\mathbf{k}), \qquad \chi = \pm 1, \tag{1}$$

with $\Delta_{x,y}$ real. The bilinears $\Delta_i^*(\mathbf{k})\Delta_j(\mathbf{k}')$ decompose into a symmetric and an antisymmetric part,

$$E \otimes E = (A_1 \oplus E)_s \oplus (A_2)_a. \tag{2}$$

For quasiparticle interference (QPI) the relevant contractions are

$$F_1(\mathbf{k},\mathbf{q}) = \Delta_x\Delta'_x + \Delta_y\Delta'_y = |\Delta||\Delta'|\cos(\phi-\phi'), \tag{3}$$
$$F_2(\mathbf{k},\mathbf{q}) = \Delta_x\Delta'_y - \Delta_y\Delta'_x = |\Delta||\Delta'|\sin(\phi-\phi'), \tag{4}$$

where primes denote $\mathbf{k}+\mathbf{q}$ and $\phi$ is the phase of $\Delta$. $F_1$ is $\mathcal{T}$-even and independent of $\chi$. $F_2 \propto \chi$ is the $A_2$ bilinear: it is odd under time reversal ($\chi \to -\chi$), odd under every vertical mirror, and invariant under their product $\sigma_v\mathcal{T}$. The chiral state therefore breaks $\mathcal{T}$ and $\sigma_v$ separately but preserves $\sigma_v\mathcal{T}$.

Note that $F_2(\mathbf{k}, \mathbf{q}\to 0) = 0$: uniform quantities never see the chirality, and an impurity (finite $\mathbf{q}$) is needed to access it.

## II. THE OBSTRUCTION: TIME REVERSAL

### A. Statement

Let $H[\chi, V]$ be the real-space Bogoliubov–de Gennes Hamiltonian including an arbitrary non-magnetic impurity potential $V$. Time reversal $\mathcal{T} = i\sigma_y K$ leaves the normal state (including Rashba SOC) and $V$ invariant and conjugates the gap, so

$$\mathcal{T}H[\chi,V]\mathcal{T}^{-1} = H[-\chi,V] \tag{5}$$

up to a global gauge transformation. Since $\mathcal{T}$ is antiunitary, it maps the retarded Green function to the advanced one. Taking the spin-summed electron element at site $\mathbf{r}$ gives

$$\rho(\mathbf{r},\omega;\chi) = \rho(\mathbf{r},\omega;-\chi) \tag{6}$$

for all $\mathbf{r}$ and $\omega$, to all orders in $V$, for any impurity shape. The same holds for the clean LDOS ($V = 0$), so the impurity-induced change $\delta\rho$ and its Fourier transform $\delta\rho(\mathbf{q},\omega)$ are even in $\chi$.

### B. How the cancellation happens

In the Born approximation, with $\hat{T} = V(\mathbf{k},\mathbf{k}+\mathbf{q})\tau_z$,

$$\delta\rho(\mathbf{q},\omega) = \mathrm{Im}\sum_{\mathbf{k}}\big[(\omega-\epsilon)(\omega-\epsilon') - F_1 - iF_2\big] \times \frac{V(\mathbf{k},\mathbf{k}+\mathbf{q})}{D(\mathbf{k})D(\mathbf{k}+\mathbf{q})}, \tag{7}$$

with $D = (\omega+i0)^2 - E^2$. Relabel $\mathbf{k} \to -\mathbf{k}-\mathbf{q}$, which exchanges the two propagators.

- The denominators are even, since $E(-\mathbf{k}) = E(\mathbf{k})$.
- $V$ is even for a non-magnetic impurity, by Hermiticity and $\mathcal{T}$.
- $F_2$ is odd, because $\Delta_x$ and $\Delta_y$ have a common parity.

The $F_2$ sum therefore vanishes. The integrand is $A_2$, but the sum is not: the QPI kernel is symmetric under exchange of the two propagators, so only the symmetric part $(E \otimes E)_s$ reaches the signal. This is Eq. (6) at first order.

### C. Mirrors

For an impurity on a mirror line, $\sigma_v : \chi \to -\chi$ combined with Eq. (6) gives $\delta\rho(\sigma_v \mathbf{r}) = \delta\rho(\mathbf{r})$. The spin-integrated map is mirror-symmetric even though the state breaks $\sigma_v$, because $\sigma_v \mathcal{T}$ is a symmetry. The $A_2$ projection of the map vanishes.

## III. THE RESOLUTION: A TIME-REVERSAL-ODD PROBE

### A. Spin-resolved LDOS

A tip with spin polarization $P$ along $\hat{\boldsymbol{m}}$ measures $dI/dV \propto \rho + P\,\hat{\boldsymbol{m}} \cdot \boldsymbol{\rho}_s$, with

$$\rho_a(\mathbf{r}, \omega) = -\frac{1}{\pi} \mathrm{Im}\, \mathrm{Tr}\big[\mathcal{P}\sigma_a \hat{G}(\mathbf{r}, \mathbf{r}; \omega)\big], \qquad a = x, y, z, \tag{8}$$

where $\mathcal{P}$ projects onto the electron block of the spinful Nambu Green function. The impurity-induced part is the usual QPI expression with $\sigma_a$ inserted into the trace.

### B. Selection rule

Since $\mathcal{T}\sigma_a\mathcal{T}^{-1} = -\sigma_a$, the argument of Sec. II now gives

$$\boldsymbol{\rho}_s(\mathbf{r}, \omega; \chi) = -\boldsymbol{\rho}_s(\mathbf{r}, \omega; -\chi) \tag{9}$$

exactly, for any non-magnetic impurity. Hence:

(a) in any $\mathcal{T}$-invariant superconductor, $\boldsymbol{\rho}_s \equiv 0$ (Kramers); a nonzero spin contrast around a non-magnetic defect, in a sample that is non-magnetic above $T_c$, proves $\mathcal{T}$ breaking;

(b) in a $\mathcal{T}$-breaking state, $\boldsymbol{\rho}_s$ is purely odd in $\chi$, and its sign images the chiral domain;

(c) reversing the tip magnetization (or comparing two domains at fixed tip) isolates $2P\boldsymbol{\rho}_s$; imperfect polarization only rescales the signal.

### C. Why SOC is needed, and how it works

For spin-singlet pairing without SOC, spin-rotation symmetry forces $\boldsymbol{\rho}_s \equiv 0$. SOC locks spin to momentum. $S_z$ transforms as $A_2$ and is $\mathcal{T}$-odd; projecting onto the Rashba helicity bands, the $\sigma_z$ insertion between $\mathbf{k}$ and $\mathbf{k}'$ produces

$$\mathrm{Tr}\big[\sigma_z P_\lambda(\mathbf{k}) P_\lambda(\mathbf{k}')\big] = \tfrac{i}{2}\,\hat{z} \cdot (\hat{\mathbf{k}} \times \hat{\mathbf{k}}'), \tag{10}$$

a pseudoscalar form factor that is odd under $\mathbf{k} \leftrightarrow \mathbf{k}'$. The relabelling $\mathbf{k} \to -\mathbf{k} - \mathbf{q}$ now cancels the even terms and keeps the odd one:

$$\delta\rho_z(\mathbf{q}, \omega) \propto \sum_{\mathbf{k}} F_2(\mathbf{k}, \mathbf{q})\, [\hat{\mathbf{k}} \times \hat{\mathbf{k}}']_z\, \mathrm{Im} \frac{V}{D(\mathbf{k}) D(\mathbf{k} + \mathbf{q})} + \dots \tag{11}$$

In symmetry language, the probe supplies a second $A_2$ factor, and $A_2 \otimes A_2 = A_1$: the chirality reaches the spin-resolved map in the fully symmetric channel. The impurity provides the momentum transfer that $F_2$ needs; the spin-polarized tip with SOC prevents its cancellation. (A complete spinful derivation, including the hole channel and the helicity gauge phases, is still to be written out; the exact statement is Eq. (9).)

## IV. SYMMETRY DECOMPOSITION

### A. Mirror parity: unitary versus antiunitary mirrors

Take a mirror $\sigma_v$ preserved by the defect.

- *Chiral states* preserve $\sigma_v\mathcal{T}$. Both $\sigma_v$ and $\mathcal{T}$ flip $S_z$, so $\rho_z(\sigma_v\mathbf{r}) = +\rho_z(\mathbf{r})$: mirror-even, like $\delta\rho$.
- *Non-chiral $\mathcal{T}$-breaking states*, e.g. $s + id_{x^2-y^2}$, preserve $\sigma_v$ itself, so $\rho_z(\sigma_v\mathbf{r}) = -\rho_z(\mathbf{r})$: mirror-odd.
- *$\mathcal{T}$-invariant states*: $\rho_z \equiv 0$.

The spin-polarized map thus reads off whether the surviving mirrors are unitary or antiunitary, i.e. the magnetic point group of the condensate.

### B. Irreducible content of the maps

Let $\Gamma_{\mathrm{imp}}$ be the representation of the defect, $\Gamma_{\mathrm{even}}$ that of the $\mathcal{T}$-even gap bilinears and $\Gamma_{\mathrm{odd}}$ that of the $\mathcal{T}$-odd ones. To leading order,

$$\delta\rho \in \Gamma_{\mathrm{imp}} \otimes \Gamma_{\mathrm{even}}, \qquad \delta\rho_z \in A_2 \otimes \Gamma_{\mathrm{odd}} \otimes \Gamma_{\mathrm{imp}} \otimes \Gamma_{\mathrm{even}}. \tag{12}$$

For a chiral state, $\Gamma_{\mathrm{even}} = A_1$ and $\Gamma_{\mathrm{odd}} = A_2$, so both maps lie in $\Gamma_{\mathrm{imp}}$: the spin-resolved map carries the same irreps as the spin-integrated one. For $s + id_{x^2-y^2}$, $\Gamma_{\mathrm{odd}} = E_2$ and the content of $\delta\rho_z$ is shifted by $A_2 \otimes E_2$.

### C. Condensate symmetry from the clean LDOS

The continuum LDOS, $\rho(\mathbf{r}) = \sum_{\mathbf{R}\mathbf{R}'} w(\mathbf{r}-\mathbf{R})w(\mathbf{r}-\mathbf{R}')\rho_{\mathbf{R}\mathbf{R}'}$, contains the bond Green functions, which carry $\Gamma_{\text{even}}$. The Bragg-peak decomposition of Ref.[3] therefore detects nematicity ($\Gamma_{\text{even}} \ni E_2$) from sub-unit-cell structure; on the triangular lattice the first Bragg shell contains $A_1 \oplus E_2$, so this is not extinct. Being $\mathcal{T}$-even, the clean LDOS cannot distinguish $d+id$ from $d-id$.

### D. Two independent tests

- clean LDOS $C_{3v}$-symmetric, $\rho_z = 0$: $\mathcal{T}$-invariant, non-nematic (e.g. extended $s$);
- clean LDOS nematic, $\rho_z = 0$: $\mathcal{T}$-invariant nematic (e.g. $d_{x^2-y^2}$, $s+d$);
- clean LDOS $C_{3v}$-symmetric, $\rho_z \neq 0$ and mirror-even: **chiral**;
- clean LDOS nematic, $\rho_z \neq 0$ and mirror-odd: non-chiral $\mathcal{T}$-breaking (e.g. $s+id$).

Once $\Gamma_{\text{even}}$ is fixed, the decomposition of $\delta\rho$ around each defect gives $\Gamma_{\text{imp}}$. The symmetry of every defect is then determined from data, and low-symmetry defects become an input rather than a failure of the method.

### E. $C_{3v}$ versus $C_{6v}$

Sn/Si(111) has $C_{3v}$ symmetry. The valley-Zeeman term ($\propto \sigma_z \sin 3\theta$) enters as $\sigma_z$ and feeds directly into $\rho_z$, so decompositions should be carried out in $C_{3v}$, where $B_1$ of $C_{6v}$ becomes $A_1$.

## V. COMPARISON WITH THE NODE–ANTINODE FINGERPRINT

*To be developed.* Wu *et al.*[1], building on Ref.[2], use the on-site LDOS of impurity bound states: at a point-like non-magnetic defect the local anomalous propagator vanishes when the gap angular momentum satisfies $\ell \not\equiv 0 \pmod{n}$, giving a node at one bound-state energy and an antinode at the other.

*Upshot.*— That condition is a statement about the irrep of the gap: the gap has no $A_1$ component, i.e. it transforms as a non-trivial irrep such as $E_2$. It is a $\mathcal{T}$-even observable, so by Eq. (6) it is blind to the $A_2$ bilinear. It can establish that the order parameter is $E_2$, but not whether the two components combine chirally ($d+id$, $A_2$ bilinear nonzero) or as a real, nematic combination ($A_2$ bilinear zero, $E_2$ bilinear nonzero). In Ref.[1] the nematic option is excluded by the shape of the tunnelling gap, not by the impurity fingerprint. Our protocol detects the $A_2$ bilinear directly, through $\rho_z$, and hence chirality.

*Planned test.*— Apply both diagnostics to two $E_2$ states with the same Fermi surface and defect: (i) chiral $d+id$; (ii) a non-chiral (nematic) $E_2$ state, e.g. real $d_{x^2-y^2}$ (optionally gapped by a weak subdominant component). To leading order the node condition holds for both, since neither gap has an $A_1$ component, so the node–antinode test cannot tell them apart. The spin-resolved signal is nonzero only for (i), as required by Eq. (9), and the clean-LDOS decomposition flags (ii) as nematic. A second test along the same lines: the node–antinode maps of $d+id$ and $d-id$ are identical by Eq. (6), whereas $\rho_z$ reverses sign.

## VI. OTHER TIME-REVERSAL-ODD PROBES

The obstruction rests on three $\mathcal{T}$-even ingredients: the normal state, the impurity and the observable. Making any one of them $\mathcal{T}$-odd can carry a chirality signal.

- *Magnetic adatom.* Time reversal gives $\rho(\chi,\lambda) = \rho(-\chi,-\lambda)$ for an impurity moment $\lambda$, to all orders. The part of the map odd in $\lambda$, $\frac{1}{2}[\delta\rho(\lambda)-\delta\rho(-\lambda)]$, is exactly odd in $\chi$; no scaling with impurity strength is needed, only reversal of the moment (not by a field, which may also retrain the domain). For a spin moment and singlet pairing this again requires SOC: a $\pi$ spin rotation maps $\lambda \to -\lambda$ without touching $\chi$.
- *Orbital moment (loop-current defect).* A vertex $V \propto i\lambda[\mathbf{k}\times\mathbf{k}']_z$ couples directly to the orbital chirality and needs no SOC; its linear-in-$\lambda$ response is $\propto \chi\lambda$. It requires a defect with an unquenched orbital moment.
- *Vortices.* The vorticity plays the role of $\lambda$; vortices are large ($\sim \xi$) and the field that creates them may switch domains.

For triplet or mixed-parity pairing the $\boldsymbol{d}$-vector already breaks spin-rotation symmetry, so spin probes can couple to $\chi$ without SOC.

## VII. OPEN ITEMS

- Full spinful Born derivation of $\delta\rho_z$ (hole channel, helicity gauge phases).
- Carry out the comparison of Sec. V.
- Practical issues for experiment: tip-field effects, magnetic defects (checked above $T_c$), setpoint normalization.

[1] X. Wu *et al.*, Microscopic fingerprint of chiral superconductivity, Phys. Rev. X (2025), DOI:10.1103/jmmf-mpr8; arXiv:2507.18693.
[2] Y. Cai and R.-X. Zhang, arXiv:2506.20842 (2025).
[3] J. Ingham, Y.-X. Jiang, M. Z. Hasan, and H. D. Scammell, Group theory method for extracting order parameters from scanning tunneling microscopy data (2026).